Problem set 2; Solutions: Geometry and gravitation

1. **Curves, scalars and covectors on the sphere** The curve considered is given, in the local chart, by:

$$\varphi \circ c(t) = \left(\frac{t}{r}, t\right) = (\theta(t), \phi(t)).$$

Hence, by definition, its tangent vector at p is:

$$\begin{aligned} X_p &= \frac{d\theta(t)}{dt}\frac{\partial}{\partial\theta}_{|\varphi(p)} + \frac{d\phi(t)}{dt}\frac{\partial}{\partial\phi}_{|\varphi(p)} \\ &= \frac{1}{r}\frac{\partial}{\partial\theta}_{|\varphi(p)} + \frac{\partial}{\partial\phi}_{|\varphi(p)}. \end{aligned}$$

This clearly defines a vector field X with $X(p) = X_p$. In the local chart, the curves tangent to X are straightlines with slope r, and when reprojected on the sphere, they are like spirals on the surface of the sphere.
With the function f given in coordinates by:

$$f \circ \varphi^{-1}(\theta, \phi) = \sin\theta\cos\phi,$$

we can define a co-vector which is its differential, or gradient, given by:

$$df_p = \frac{\partial f}{\partial\theta}d\theta + \frac{\partial f}{\partial\phi}d\phi.$$

The result follows.
Given the one-form ω_p:

$$\omega_p = -d\theta + d\phi,$$

we get, by definition:

$$\omega_p(X_p) = \omega_{pi}X_p^i = -\frac{1}{r} + 1.$$

2. **Real projective space**
If you are not comfortable with this, review what an equivalence relation is.
In order to show that the coordinates $\xi_{(i)}$ do not depend on the choice of representative of the elements of $\mathbb{R}P^2$, choose two such representatives of a line p, (x^0, x^1, x^2) and (y^0, y^1, y^2). Then, we know that there exists a non-zero real number λ such that $x^j = \lambda y^j$. Suppose that $x^0 \neq 0$ (the reasoning is the same in the other cases). Then, necessarily, $y^0 \neq 0$. Besides, $p \in U_0$, and we have:

$$\xi_{(0)}(p) = \left(\frac{x^1}{x^0}, \frac{x^2}{x^0}\right) = \left(\frac{\lambda x^1}{\lambda x^0}, \frac{\lambda x^2}{\lambda x^0}\right) = \left(\frac{y^1}{y^0}, \frac{y^2}{y^0}\right).$$

Therefore, the local coordinates on $\mathbb{R}P^2$ do not depend on the representative of p chosen.
The coordinates $\xi_{(i)}$ are clearly homeomorphic; their inverses are given by simple maps:

$$\begin{aligned}\xi_{(0)}^{-1}(u,v) &= (1,u,v)\\ \xi_{(1)}^{-1}(u,v) &= (u,1,v)\\ \xi_{(2)}^{(-1)}(u,v) &= (u,v,1)\ ,\end{aligned}$$

where we chose representative such that the non-zero component in the triplets $\left(x^0,x^1,x^2\right)$ is 1, for simplicity. The transition functions are then obtained very simply; for example, ψ_{10} is given by:

$$\psi_{10}(u,v)=\left(\frac{1}{u},\frac{v}{u}\right)\ ,$$

with a domain which is $u\in\mathbb{R}^*$ and $v\in\mathbb{R}$; so they are clearly smooth.
The identification with the half-sphere is simple and almost obvious using a figure.

3. **Spin group as a manifold**
Relatively pedestrian. You can find that in any good differential geometry textbook.

4. **Two charts on the sphere**
Pedestrian calculations

5. **Vector fields in the plane**
Integral curves of $\boldsymbol{X}$ with parameter λ are determined by:

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}\lambda}&=x\\ \frac{\mathrm{d}y}{\mathrm{d}\lambda}&=-y\ .\end{aligned}$$

Therefore, we get $x(\lambda)=x_0+\mathrm{e}^{\lambda}$ and $y(\lambda)=y_0+\mathrm{e}^{-\lambda}$ for arbitrary constants x_0 and y_0. Thus:

$$y=y_0+\frac{1}{x-x_0}\ .$$

These are hyperbolæ. Similarly, the integral curves of $\boldsymbol{Y}$ are straight lines.
To determine the commutator, let us use an arbitrary function f. Then:

$$\begin{aligned}[\boldsymbol{X},\boldsymbol{Y}]\,f=&\boldsymbol{X}\left(\boldsymbol{Y}(f)\right)-\boldsymbol{Y}\left(\boldsymbol{X}(f)\right)\\ =&\boldsymbol{X}\left(x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}\right)-\boldsymbol{Y}\left(x\frac{\partial f}{\partial x}-y\frac{\partial f}{\partial y}\right)\\ =&x\frac{\partial}{\partial x}\left(x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}\right)-y\frac{\partial}{\partial y}\left(x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}\right)-x\frac{\partial}{\partial x}\left(x\frac{\partial f}{\partial x}-y\frac{\partial f}{\partial y}\right)-y\frac{\partial}{\partial y}\left(x\frac{\partial f}{\partial x}-y\frac{\partial f}{\partial y}\right)\\ =&0\ .\end{aligned}$$

6. **Conformal transformations**
Lightlike curves are characterised by there tangent vector $\boldsymbol{k}$, which is lightlike at every point p on the curve:

$$\boldsymbol{g}\left(\boldsymbol{k},\boldsymbol{k}\right)=g_{\mu\nu}k^{\mu}k^{\nu}=0\ .$$

Let $\tilde{\boldsymbol{g}} = f\boldsymbol{g}$ be the new metric. Then, clearly, for the same curve, with the same tangent vector at each point:

$$\tilde{\boldsymbol{g}}\left(\boldsymbol{k},\boldsymbol{k}\right) = \tilde{g}_{\mu\nu}k^{\mu}k^{\nu} = f\left(x^{\alpha}\right)g_{\mu\nu}k^{\mu}k^{\nu} = 0\ ,$$

which shows that the curve remains lightlike.
For timelike curves, the tangent vector satisfies:

$$\boldsymbol{g}\left(\boldsymbol{k},\boldsymbol{k}\right) = g_{\mu\nu}k^{\mu}k^{\nu} < 0\ .$$

This can only be preserved under a conformal transformation if f is a strictly positive function.

7. **Local Lorentz factor and observables**
   See notes.

8. **Affine connections**
   See notes.

9. **Index manipulation etc.**
   The key is to prove the first expression and this requires one to know the differential of the determinant. For invertible matrices with real coefficients, this is actually not that hard. Let us start with the differential at the identity. Let $\boldsymbol{I}$ be the identity and $\boldsymbol{X}$ a real square matrix in dimension n whose (complex) eigenvalues are denoted λ_i (with possible repetitions). Then (using the characteristic polynomial of $\boldsymbol{M}$) for $t \in \mathbb{R}$:

$$\begin{aligned}\det\left[\boldsymbol{I} + t\boldsymbol{M}\right] =& \Pi_{i=1}^{n}\left(1 + t\lambda_i\right)\\ =& 1 + t\mathrm{Tr}\left(\boldsymbol{M}\right) + O\left(t^2\right)\ .\end{aligned}$$

Therefore:

$$D_{\boldsymbol{I}}\det(\boldsymbol{M}) = \mathrm{Tr}\left(\boldsymbol{M}\right)\ .$$

To get the differential away from the identity but for an arbitrary invertible matrix $\boldsymbol{M}$ we write for an arbitrary square matrix $\boldsymbol{H}$:

$$\begin{aligned}\det\left[\boldsymbol{M} + \boldsymbol{H}\right] =& \det\left[\boldsymbol{M}\left(\boldsymbol{I} + \boldsymbol{M}^{-1}\boldsymbol{H}\right)\right]\\ =& \det\left[\boldsymbol{M}\right]\left(1 + \mathrm{Tr}\left(\boldsymbol{M}^{-1}\boldsymbol{H}\right) + O\left(\|\boldsymbol{H}\|^2\right)\right)\\ \simeq& \det\left[\boldsymbol{M}\right] + \mathrm{Tr}\left(\det(\boldsymbol{M})\boldsymbol{M}^{-1}\boldsymbol{H}\right)\ .\end{aligned}$$

Therefore[1]:

$$D_{\boldsymbol{M}}\det(\boldsymbol{H}) = \det(\boldsymbol{M})\mathrm{Tr}\left(\boldsymbol{M}^{-1}\boldsymbol{H}\right)\ .$$

[1] Using the fact that invertible matrices are dense in the set of square matrices, this can be generalised to arbitrary square matrices by introducing the comatrix of $\boldsymbol{M}$.

Applying that to the metric, we get:

$$\begin{aligned}
\det\left[g_{\mu\nu}\left(x^\beta+\delta x^\beta\right)\right] \simeq & \det\left[g_{\mu\nu}\left(x^\beta\right)+\underbrace{\partial_\alpha g_{\mu\nu}\delta x^\alpha}_{=H_{\mu\nu}}\right] \\
\simeq & \det\left[g_{\mu\nu}\left(x^\beta\right)\right]+\det\left[g_{\mu\nu}\left(x^\beta\right)\right]\mathrm{Tr}\left(g^{\mu\rho}H_{\rho\nu}\right) \\
\simeq & g\left(x^\beta\right)+gg^{\mu\nu}H_{\mu\nu} \\
\simeq & g\left(x^\beta\right)+gg^{\mu\nu}\partial_\alpha g_{\mu\nu}\delta x^\alpha \ .
\end{aligned}$$

Thus:

$$\partial_\alpha g = gg^{\mu\nu}\partial_\alpha g_{\mu\nu} \ .$$

The other results follow by calculations.

10. **Parallel transport on the sphere**
See handwritten solutions attached.

11. **The cylinder is flat**
Draw!

12. **2D de Sitter space**
See handwritten solutions attached.

13. **Lie derivative: an exercise**
The first relation follows directly from the definition. The Leibniz rule follows from the fact that it is obeyed by vectors (you can see it by coming back to coordinates if you are not convinced). For an arbitrary function g:

$$\begin{aligned}
\mathcal{L}_{\boldsymbol{u}}\left(f\boldsymbol{v}\right)g = & \left[\boldsymbol{u},f\boldsymbol{v}\right]g \\
= & \boldsymbol{u}\left(f\boldsymbol{v}(g)\right)-f\boldsymbol{v}\left(\boldsymbol{u}(g)\right) \\
= & \boldsymbol{u}(f)\boldsymbol{v}(g)+f\boldsymbol{u}\left(\boldsymbol{v}(g)\right)-f\boldsymbol{v}\left(\boldsymbol{u}(g)\right) \\
= & \left(\mathcal{L}_{\boldsymbol{u}}f+f\mathcal{L}_{\boldsymbol{u}}\boldsymbol{v}\right)g \ .
\end{aligned}$$